From LED Activation to Photon Flux

Irradiance Computation Rationale

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1 Overview

This document describes the method to convert a white LED source activation level x (%) into physically meaningful irradiance quantities: power $P(x)$ [mW cm^{-2}] and photon flux $\Phi(x)$ [photons cm^{-2} s^{-1}]. The computation is implemented in `light_irradiance.py` and relies on two inputs:

1. A **power look-up table** $\{x_i \to P(x_i)\}$, obtained from direct irradiance measurements at discrete Xcite activation levels (Table 1).
2. A **normalised spectral shape** $I(\lambda)$ [a.u.], measured once and stored in `white_mea1.pkl`.

2 Key Assumption

The spectral *shape* of the white LED source is assumed invariant with activation level; only the *amplitude* scales with x:

$$L(\lambda, x) = S(x) \cdot I(\lambda), \tag{1}$$

where $S(x)$ [mW cm^{-2} nm^{-1}] is the peak spectral irradiance at activation x, and $I(\lambda)$ is the normalised spectrum (peak = 1, in a.u.). This is a standard and well-validated approximation for LED sources; the normalised spectrum $I(\lambda)$ is shown in Figure 1 (left).

3 Derivation

3.1 Spectral irradiance $L(\lambda, x)$

By definition, integrating the spectral irradiance over wavelength recovers the total power:

$$P(x)\ [\mathrm{mW\,cm^{-2}}] = \int L(\lambda, x)\,\mathrm{d}\lambda. \tag{2}$$

Substituting Equation (1) into (2):

$$P(x) = S(x) \int I(\lambda)\,\mathrm{d}\lambda = S(x) \cdot I_{\mathrm{tot}}, \tag{3}$$

where $I_{\mathrm{tot}} = \int I(\lambda)\,\mathrm{d}\lambda$ [nm] is the area under the normalised spectrum. Solving for $S(x)$:

$$\boxed{S(x) = \frac{P(x)}{I_{\mathrm{tot}}} = \frac{P(x)}{\displaystyle\int I(\lambda)\,\mathrm{d}\lambda}} \quad [\mathrm{mW\,cm^{-2}\,nm^{-1}}]. \tag{4}$$

The full spectral irradiance at any activation level x is therefore:

$$\boxed{L(\lambda, x) = I(\lambda) \cdot \frac{P(x)}{I_{\text{tot}}}} \quad [\text{mW cm}^{-2}\,\text{nm}^{-1}]. \tag{5}$$

$P(x)$ for arbitrary x (not necessarily in the look-up table) is obtained by linear interpolation of the calibration measurements (Section 4).

3.2 Power $P(x)$ — self-consistency check

Equation (2) can be used as a self-consistency check: integrating $L(\lambda, x)$ back over λ must recover the measured (or interpolated) $P(x)$:

$$P(x) = \int L(\lambda, x)\,\mathrm{d}\lambda = \frac{P(x)}{I_{\text{tot}}} \int I(\lambda)\,\mathrm{d}\lambda = P(x). \quad \checkmark \tag{6}$$

This check is implemented as an assertion in the `__main__` block of `light_irradiance.py`.

3.3 Photon flux $\Phi(x)$

The energy of a single photon at wavelength λ is:

$$E_{\text{photon}}(\lambda) = \frac{hc}{\lambda} \quad [\text{J}], \tag{7}$$

with $h = 6.626 \times 10^{-34}$ J · s (Planck constant) and $c = 2.998 \times 10^{17}$ nm s^{-1} (speed of light). The spectral photon flux density [photons s^{-1} cm^{-2} nm^{-1}] is:

$$\phi(\lambda, x) = \frac{L(\lambda, x)}{E_{\text{photon}}(\lambda)} = \frac{L(\lambda, x) \cdot \lambda}{hc}, \tag{8}$$

where L is expressed in [mW cm^{-2} nm^{-1}] after conversion to [J s^{-1} cm^{-2} nm^{-1}] via the factor 10^{-3}. Integrating over wavelength:

$$\boxed{\Phi(x) = \int \frac{L(\lambda, x) \cdot \lambda}{hc}\,\mathrm{d}\lambda \times 10^{-3}} \quad [\text{photons cm}^{-2}\,\text{s}^{-1}]. \tag{9}$$

All integrals are evaluated numerically with the trapezoidal rule (`numpy.trapz`), which handles non-uniform λ spacing.

4 Power Look-Up Table and Interpolation

Table 1 lists the calibration measurements of the white LED source (MEA1, no ND filter, Awen 23.06.2025).

For activation levels not in the table, $P(x)$ is obtained by linear interpolation (`numpy.interp`); out-of-range values return `NaN` and trigger an assertion error. The function `_interpolate_power_lookup` in `light_irradiance.py` implements this step.

5 Implementation Summary

The pipeline is implemented in `light_irradiance.py` and proceeds as follows:

1. **Load** the normalised spectrum $I(\lambda)$ from `white_mea1.pkl` via `_get_spectrum()`.

Table 1: Power look-up table `SOURCE_PTG_POWER_mW_cm2`.

Xcite level x (%)	$P(x)$ [mW cm^{-2}]
1	0.345
10	3.2
20	6.9
30	10.5
40	14.0
50	17.4
60	20.9
70	24.7
80	28.3
90	32.3
100	37.7

2. **Interpolate** $P(x)$ from the look-up table via `_interpolate_power_lookup()`.
3. **Compute** $L(\lambda, x)$ via `compute_spectral_irradiance()` using Equation (5).
4. **Integrate** $L(\lambda, x)$ to obtain either $P(x)$ or $\Phi(x)$ via `compute_irradiance()` with `unit='power_mW_cm2'` or `unit='photon_flux_cm2_s'`.

6 Results

Figure 1 shows the normalised white LED spectrum (left) and the resulting spectral irradiance $L(\lambda, x)$ for several activation levels (right). Figure 2 shows the self-consistency check (computed vs. measured power, left) and the photon flux as a function of activation level (right). Both quantities scale linearly with x, as expected from Equations (4)–(9).

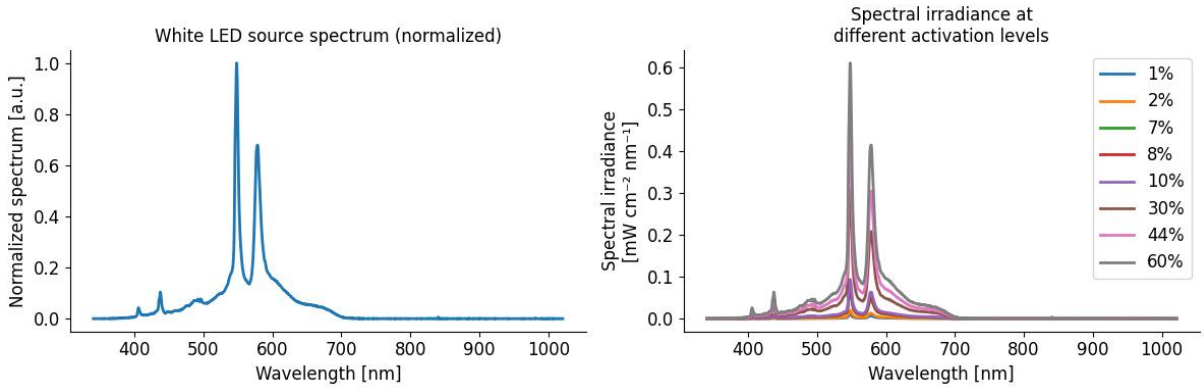

Figure 1: **Left:** Normalised spectrum $I(\lambda)$ of the white LED source (MEA1). **Right:** Spectral irradiance $L(\lambda, x)$ at different Xcite activation levels, computed from Equation (5).

Files

- `light_irradiance.py` — full implementation
- `white_mea1.pkl` — normalised spectrum $I(\lambda)$
- `white_mea1.jpg` — Figure 1

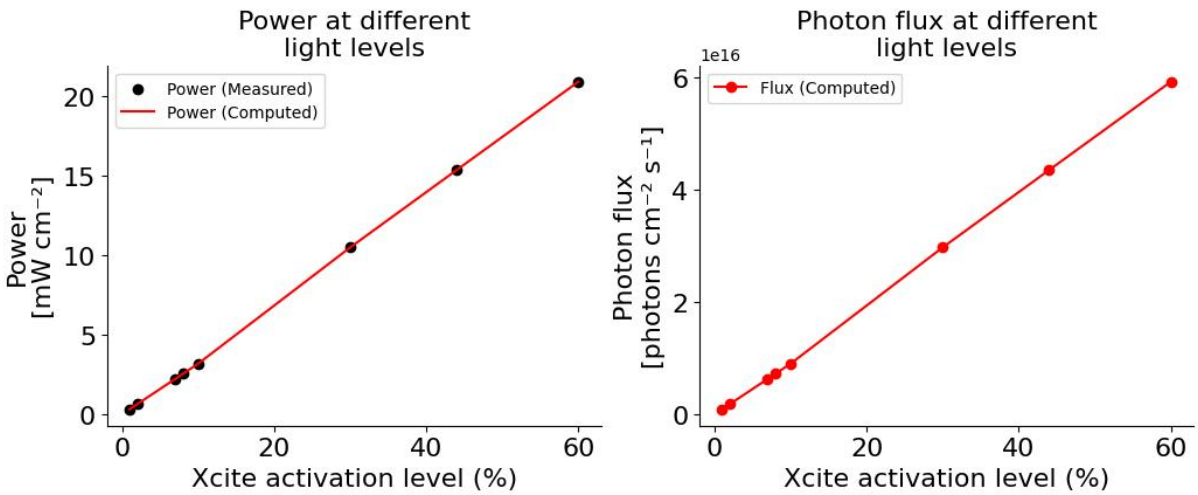

Figure 2: **Left:** Self-consistency check — power recovered by integrating $L(\lambda, x)$ (red line) agrees with interpolated measurements (black dots). **Right:** Photon flux $\Phi(x)$ computed from Equation (9).

- `white_mea1_power_flux_computation.jpg` — Figure 2
- `irradiance_computation.pdf` — this document (compiled)